Grokking as the Transition from Lazy to Rich Training Dynamics

Exercise Session — 1h, pen and paper

Reference: Kumar, Bordelon, Gershman & Pehlevan, ICLR 2024

Problem 1 — The Model and its Summary Statistics (10 min)

Consider a two-layer committee machine with N hidden neurons, input $\boldsymbol{x} \in \mathbb{R}^D$, and fixed readout weights all equal to 1:

$$f(\boldsymbol{w}, \boldsymbol{x}) = \frac{\alpha}{N} \sum_{i=1}^{N} \varphi(w_i \cdot \boldsymbol{x}), \qquad \varphi(h) = h + \frac{\epsilon}{2} h^2$$

The target function is $y(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{\beta}_\star \cdot \boldsymbol{x})^2$, with $\boldsymbol{x} \sim \mathcal{N}(0, \frac{1}{D}\boldsymbol{I})$.

(a) Define the two summary statistics:

$$\bar{\boldsymbol{w}} = \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{w}_i, \qquad \boldsymbol{M} = \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{w}_i \boldsymbol{w}_i^\top$$

Show that the network output can be written as:

$$f(\boldsymbol{x}) = \alpha\, \bar{\boldsymbol{w}} \cdot \boldsymbol{x} \; + \; \frac{\alpha\epsilon}{2}\, \boldsymbol{x}^\top \boldsymbol{M}\, \boldsymbol{x}$$

(b) What are the values of $\bar{\boldsymbol{w}}$ and $\boldsymbol{M}$ at random initialisation (large N limit)?

(c) For the network to perfectly fit the target on the test distribution, what must $\boldsymbol{M}$ and $\bar{\boldsymbol{w}}$ equal? (*Hint*: match the quadratic and linear parts of f to y.)

Problem 3 — The NTK of the Toy Model (10 min)

(a) Compute the NTK $K(\boldsymbol{x}, \boldsymbol{x}') = \sum_{i=1}^{N} \nabla_{w_i} f \cdot \nabla_{w_i} f$ for this model. Show that it can be expressed in terms of $\bar{\boldsymbol{w}}$ and $\boldsymbol{M}$ as:

$$K(\boldsymbol{x}, \boldsymbol{x}') = (\boldsymbol{x} \cdot \boldsymbol{x}') + \epsilon\, (\boldsymbol{x} \cdot \boldsymbol{x}')\, \bar{\boldsymbol{w}} \cdot (\boldsymbol{x} + \boldsymbol{x}') + \epsilon^2\, (\boldsymbol{x} \cdot \boldsymbol{x}')\, \boldsymbol{x}^\top \boldsymbol{M}\, \boldsymbol{x}'$$

(*Hint*: compute $\nabla_{w_i} f$ first, then form the dot product.)

(b) At initialisation ($\bar{\boldsymbol{w}} = 0$, $\boldsymbol{M} = \boldsymbol{I}$), the kernel simplifies to $K_0(\boldsymbol{x}, \boldsymbol{x}') = (\boldsymbol{x} \cdot \boldsymbol{x}') + \epsilon^2 (\boldsymbol{x} \cdot \boldsymbol{x}')^2$. This kernel is a sum of two terms. The first term $(\boldsymbol{x} \cdot \boldsymbol{x}')$ is

sensitive to *linear* structure in the data, and the second $\epsilon^2(\boldsymbol{x}\cdot\boldsymbol{x}')^2$ to *quadratic* structure. Compute the ratio of the typical magnitude of the quadratic term to the linear term when x, x' are independent draws from $\mathcal{N}(0, \frac{1}{D}I)$. What does this tell you about how well-suited the initial kernel is for learning the (purely quadratic) target, especially when ϵ is small?

(*Hint*: use $\mathbb{E}[(\boldsymbol{x}\cdot\boldsymbol{x}')^2] = 1/D$ and $\mathbb{E}[(\boldsymbol{x}\cdot\boldsymbol{x}')^4] \sim 1/D^2$ for independent draws.)

(c) The target $y(x) = \frac{1}{2}(\beta_\star \cdot x)^2$ is purely quadratic. After feature learning, the network can align $\boldsymbol{M}$ with $\beta_\star\beta_\star^\top$, effectively reducing the problem to learning a single direction in $\mathbb{R}^D$. Argue intuitively that this requires $P \sim D$ samples (think about how many equations you need to identify a vector in $\mathbb{R}^D$). By contrast, a fixed kernel that treats all $\sim D^2$ quadratic directions equally would need far more samples. Why does this gap between D and D^2 create an opportunity for grokking?

Problem 4 — Loss Decomposition (15 min)

The test MSE of the model is $\mathcal{L} = \langle (y-f)^2\rangle$ where $\langle\cdot\rangle$ is the expectation over $x \sim \mathcal{N}(0, \frac{1}{D}I)$.

Useful identity (Isserlis' theorem for Gaussians). For $x \sim \mathcal{N}(0, \frac{1}{D}I)$ and any symmetric matrices A, B:

$$\langle (x^\top A\, x)(x^\top B\, x)\rangle = \frac{1}{D^2}\left[(\operatorname{Tr} A)(\operatorname{Tr} B) + 2\operatorname{Tr}(AB)\right]$$

In particular, $\langle (x^\top A\, x)^2\rangle = \frac{1}{D^2}[(\operatorname{Tr} A)^2 + 2|A|_F^2]$ where $|A|_F^2 = \operatorname{Tr}(A^\top A)$.

(a) Decompose the network output as $f = f_{\text{lin}} + f_{\text{quad}}$ with $f_{\text{lin}} = \alpha\bar{\boldsymbol{w}}\cdot\boldsymbol{x}$ and $f_{\text{quad}} = \frac{\alpha\epsilon}{2}\boldsymbol{x}^\top\boldsymbol{M}\,\boldsymbol{x}$. Note that the target $y = \frac{1}{2}\boldsymbol{x}^\top\boldsymbol{\beta}_\star\boldsymbol{\beta}_\star^\top\boldsymbol{x}$ is purely quadratic. Use the fact that odd moments of isotropic Gaussians vanish (i.e. $\langle x_i\, g(x^\top Ax)\rangle = 0$) to argue that the MSE splits as:

$$\mathcal{L} = \langle (y - f_{\text{quad}})^2\rangle + \langle f_{\text{lin}}^2\rangle$$

Then compute $\langle f_{\text{lin}}^2\rangle$ explicitly. This is **Term C** (linear power).

(b) Now focus on the quadratic part. Define $A = \boldsymbol{\beta}_\star\boldsymbol{\beta}_\star^\top - \alpha\epsilon\,\boldsymbol{M}$, so that $y - f_{\text{quad}} = \frac{1}{2}\boldsymbol{x}^\top A\,\boldsymbol{x}$. Apply the Isserlis identity above to compute $\langle (y-f_{\text{quad}})^2\rangle = \frac{1}{4}\langle(\boldsymbol{x}^\top A\,\boldsymbol{x})^2\rangle$. Show that the result can be written as the sum of two terms:

$$\underbrace{\frac{1}{4D^2}(\operatorname{Tr} A)^2}_{\text{Term A: variance error}} + \underbrace{\frac{1}{2D^2}|A|_F^2}_{\text{Term B: misalignment error}}$$

Write out Term A and Term B explicitly in terms of $|\boldsymbol{\beta}_\star|^2$, $\operatorname{Tr}\boldsymbol{M}$, α, ϵ, and $|\alpha\epsilon\boldsymbol{M} - \boldsymbol{\beta}_\star\boldsymbol{\beta}_\star^\top|_F$.

(c) Interpret each of the three terms (A, B, C) in one sentence.

(d) At initialisation ($\bar{\boldsymbol{w}} = 0$, $\boldsymbol{M} = \boldsymbol{I}$), which terms are large and which are small? What happens in early training: recalling from Problem 3(b) that the initial kernel is much more sensitive to linear structure than quadratic structure, which component of f does the network use first to reduce train loss, and what does this do to the three terms?

Problem 5 — The Role of α: Controlling Laziness (10 min)

(a) Consider training with gradient descent on MSE loss $\mathcal{L} = \frac{1}{2}\sum_\mu (y_\mu - f(\boldsymbol{w}, \boldsymbol{x}_\mu))^2$. In the lazy regime, f is linear in $\boldsymbol{w}$, so the gradient descent dynamics become $\dot{\boldsymbol{w}} = -\nabla_w \mathcal{L}$. For a linear model $f_\mu \approx f_\mu^0 + \boldsymbol{J}_\mu \cdot (\boldsymbol{w} - \boldsymbol{w}_0)$ where $\boldsymbol{J}_\mu = \nabla_w f|_{w_0}(\boldsymbol{x}_\mu)$, the converged solution satisfies $f_\mu = y_\mu$ for all training points. Show that the required parameter displacement is:

$$\boldsymbol{w}^* - \boldsymbol{w}_0 = \boldsymbol{J}^\top (K_0)^{-1}(\boldsymbol{y} - \boldsymbol{f}^0)$$

where $(K_0)_{\mu\nu} = \boldsymbol{J}_\mu \cdot \boldsymbol{J}_\nu$ is the NTK Gram matrix. Deduce that $|\boldsymbol{w}^* - \boldsymbol{w}_0|^2 = (\boldsymbol{y} - \boldsymbol{f}^0)^\top K_0^{-1}(\boldsymbol{y} - \boldsymbol{f}^0)$.

(*Hint*: the converged condition is $\boldsymbol{J}_\mu \cdot (\boldsymbol{w}^* - \boldsymbol{w}_0) = y_\mu - f_\mu^0$ for all μ. This is a linear system.)

(b) Now recall that the network output is $f = \frac{\alpha}{N}\sum_i \varphi(w_i \cdot x)$. A small parameter change δw produces a change in the output $\delta f \sim \alpha\,|\delta w|$ (because the Jacobian $\nabla_w f$ scales as α). Since the target is $O(1)$, the network needs $\delta f \sim O(1)$ to fit the data. Use this to argue that $|\delta w| \sim 1/\alpha$. Why does this mean that large α keeps the network in the lazy regime?

(c) Feature learning corresponds to the *nonlinear* part of the dynamics — the deviation of $f(\boldsymbol{w})$ from its linearisation. One can show that the rate at which the NTK itself changes (i.e. the rate at which features evolve) scales as $\dot{K}/K \sim |\delta w|^2 \sim 1/\alpha^2$. Meanwhile, the kernel regression solution is reached on a timescale that does not grow with α. This creates a separation of timescales.

On the diagram below, fill in the three labels (i), (ii), (iii) and mark the grokking gap. Then sketch a second curve (dashed) showing what happens when α is *increased*.

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Loss
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 |  -----------
 | /           \-----------
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|/                          \
|                            \_____________
|
+------+----------+-----------+----------→ log(time)
      (i)        (ii)       (iii)

(i)   = ...................
(ii)  = ...................
(iii) = ...................

Grokking gap = from ......... to .........
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(*Hint*: one curve is train loss, one is test loss. They start together, then diverge, then reconverge.)

(d) What happens in the limits $\alpha \to 0$ and $\alpha \to \infty$?

Problem 6 — The Role of ϵ: Initial Kernel Alignment (5 min)

(a) From Problem 3, the initial kernel eigenvalue on quadratic functions is $\lambda_{\text{quad}} = 2\epsilon^2/D^2$. Define the *NTK alignment* $\varepsilon = \boldsymbol{y}^\top K_0 \boldsymbol{y}/(\|K_0\|_F \|\boldsymbol{y}\|^2)$. How does this alignment scale with ϵ in the toy model?

(b) Large ϵ means the initial kernel already places significant power on quadratic functions. Explain why this *reduces* or *eliminates* grokking. Conversely, why does small ϵ *increase* the grokking delay but lead to *lower* final test loss?

Problem 7 — Synthesis: Why Grokking is a Lazy-to-Rich Transition (5 min)

(a) State the three conditions identified by Kumar et al. that are jointly sufficient for grokking.

(b) Using the loss decomposition from Problem 4, describe the two-phase dynamics during grokking: - Phase 1 (early training): What happens to Terms A, B, C? - Phase 2 (late training): What happens to Terms A, B, C?

(c) Explain in 2–3 sentences why weight decay is *sufficient but not necessary* for grokking, and how the lazy-to-rich framework subsumes weight-decay-based explanations.